

FitTrack AI

Personalized Fitness Recommendations Powered by AI

Phase 8 – Project Demonstration

Project type: AI-powered fitness tracking and recommendation REST API / backend

Stack: Node.js · Express.js · MongoDB Atlas · Mongoose · JWT · bcrypt.js · Google Gemini AI · Thunder Client / Postman

This document is the plan and script for the demonstration. It contains no screenshots, API responses, tokens, keys, credentials, test results or video URL. Those must come from your working project. Adjust the narration to what actually appears on screen when you record.

1. Demo Plan

A. Project Name

FitTrack AI – Personalized Fitness Recommendations Powered by AI

B. Project Purpose

FitTrack AI is a REST API backend that lets users keep a secure record of their workouts and get AI-generated help. Registered users log in with JWT authentication, manage and search their own workouts stored in MongoDB Atlas, and request a workout recommendation and fitness insights from Google Gemini AI.

C. Key Benefits

Secure workout management	Users can add, view, update and delete workouts, and only after logging in.
Organized workout history	Workouts are stored in a structured form (name, category, duration, calories, date) instead of scattered notes.
Workout search	Past workouts can be found quickly with a search query.

Personalized AI recommendations	Gemini AI suggests a workout based on the user's age, fitness goal and experience.
AI-generated fitness insights	Gemini AI comments on the user's total workouts, average duration and total calories burned.
Protected user-specific data	Passwords are stored as bcrypt hashes, routes are protected with JWT, and each workout is linked to its owner.
REST API based architecture	Any client that can send HTTP requests can use the backend, and the layers are modular and easy to extend.

D. Technology Stack

Node.js	Server-side JavaScript runtime
Express.js	Web framework for routes, middleware and controllers
MongoDB Atlas	Cloud database for users and workouts
Mongoose	Schemas, validation and the Workout → User reference
JWT	Token-based authentication
bcrypt.js	Password hashing
Google Gemini AI	AI recommendations and insights
Thunder Client / Postman	Sending API requests during the demonstration

E. Demonstration Objectives

The demonstration shows:

1. User authentication (registration, login, protected profile)
2. Workout management (add, list, view by ID, update, delete)
3. Workout search
4. AI workout recommendation
5. AI fitness insights
6. Error and validation handling
7. The final API output

2. Demo Video Script

Target length: about 3–5 minutes (the script below runs about 4 minutes 35 seconds when read at a steady pace). Timings are guides; shorten the narration if you run over.

2.1 Before Recording

- Start the server and confirm the MongoDB Atlas connection works.
- Prepare demo-only data: a dedicated demo name, email and password. Do not use a personal password.
- Prepare Thunder Client / Postman requests in the order of the scenes, with a `token` environment variable for the JWT.
- Keep the architecture diagram ready to display.
- Do not show the `.env` file, the JWT, the MongoDB connection string or the Gemini API key. Crop or blur if they appear.
- Increase the font/zoom so the request and response are readable, and close unrelated windows and notifications.
- Plan two demo workouts in Scene 7: delete only one of them, so an existing workout remains for the search in Scene 8.

2.2 Demo Sequence Summary

2.3 Scene-by-Scene Script

3. Output Evidence

Capture these screenshots from the working project (server terminal and Thunder Client / Postman). Each one should clearly show the request (method, URL, body where relevant) and the response with its status code. Do not use mock-ups. Hide or blur any JWT, credentials and API keys.

4. Demo Video Information

Project Name	FitTrack AI – Personalized Fitness Recommendations Powered by AI
Demo Video Platform	Google Drive
Video Link	[Google Drive Demo Video Link – To be added]
Access	Anyone with the link – Viewer
Video Duration	Approximately 3–5 minutes
Video Contents	• Project introduction • Purpose • Benefits • Architecture • Working process • Authentication • Workout management • Search • AI recommendation • AI fitness insights • Validation / error handling • Final output

Demonstration Flow Diagram

FitTrack AI – Project Demonstration Flow Introduction Authentication Workouts Gemini AI Error handling Final output Read left to right on row 1, right to left on row 2, and left to right on row 3. Figure 1: FitTrack AI – Project Demonstration Flow

Presentation Checklist

Phase 8 Conclusion

The demonstration presents the completed FitTrack AI system: its working process from registration and login to workout management and search, its major features, its AI capabilities (Gemini-powered recommendations and fitness insights), and the final API outputs. It also shows how validation and error handling protect the API. The plan, video script, evidence checklist and video information in this document make sure the demonstration is complete and the evidence comes from the actual working project. The demo video link and the screenshots are to be added after recording.